Customer Behavior Analysis

Internship Project

By: Rudrashetty Nandini

Close

```
#🛍️ Customer Behavior Analysis

## Data Analytics Internship Project

### 👩‍💻 Submitted By
### Rudrashetty Nandini

### 🎯 Project Objective
The objective of this project is to analy
behavior, spending patterns, and engageme
analytics and visualization techniques.

## 🛠️ Tools & Technologies Used
- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

## 📌 Key Analysis Performed
- Data Cleaning & Preprocessing
- Customer Spending Analysis
- Age Group Analysis
- Purchase Trend Analysis
- Customer Segmentation
- Visualization & Insights

## 📈 Expected Outcome
The analysis helps identify customer pref
and business opportunities for improving
sales performance.
```

🛍️ Customer Behavior Analysis

Data Analytics Internship Project

👩‍💻 Submitted By

Rudrashetty Nandini

🎯 Project Objective

The objective of this project is to analyze customer purchasing behavior, spending patterns, and engagement trends using data analytics and visualization techniques.

🛠️ Tools & Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

📌 Key Analysis Performed

- Data Cleaning & Preprocessing
- Customer Spending Analysis
- Age Group Analysis
- Purchase Trend Analysis
- Customer Segmentation
- Visualization & Insights

📈 Expected Outcome

The analysis helps identify customer preferences, spending behavior, and business opportunities for improving customer engagement and sales performance.

📈 Customer Behavior Analysis

```
from google.colab import files

uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving ecommerce_customer_data_custom_ratios.csv to ecommerce_customer_data_c
```

📁 Import Libraries

```
import os

os.listdir('/content')
```

```
['.config',
 'ecommerce_customer_data_large.csv',
 'ecommerce_customer_data_custom_ratios (1).csv',
 '.ipynb_checkpoints',
 'ecommerce_customer_data_custom_ratios.csv',
 'ecommerce_customer_data_large (1).csv',
 'sample_data']
```

📁 Load Dataset

```
import pandas as pd

df1 = pd.read_csv('/content/ecommerce_customer_data_large.csv')
```

```
df2 = pd.read_csv('/content/ecommerce_customer_data_custom_ratios.csv')

print(df1.columns)
print(df2.columns)
```

```
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
       'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',
       'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
      dtype='object')
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
       'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',
       'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
      dtype='object')
```

```
print(df1.head())

print(df2.head())
```

```
   Customer ID        Purchase Date Product Category  Product Price  Quantity
0        44605  2023-05-03 21:30:02             Home            177         1
1        44605  2021-05-16 13:57:44      Electronics            174         3
2        44605  2020-07-13 06:16:57            Books            413         1
3        44605  2023-01-17 13:14:36      Electronics            396         3
4        44605  2021-05-01 11:29:27            Books            259         4

   Total Purchase Amount Payment Method  Customer Age  Returns Customer Name
0                   2427         PayPal            31      1.0   John Rivera
1                   2448         PayPal            31      1.0   John Rivera
2                   2345    Credit Card            31      1.0   John Rivera
3                    937           Cash            31      0.0   John Rivera
4                   2598         PayPal            31      1.0   John Rivera

   Age  Gender  Churn
0   31  Female      0
1   31  Female      0
2   31  Female      0
3   31  Female      0
4   31  Female      0
   Customer ID        Purchase Date Product Category  Product Price  Quantity
0        46251  2020-09-08 09:38:32      Electronics             12         3
1        46251  2022-03-05 12:56:35             Home            468         4
2        46251  2022-05-23 18:18:01             Home            288         2
3        46251  2020-11-12 13:13:29         Clothing            196         1
4        13593  2020-11-27 17:55:11             Home            449         1

   Total Purchase Amount Payment Method  Customer Age  Returns  \
0                    740    Credit Card            37      0.0
1                   2739         PayPal            37      0.0
2                   3196         PayPal            37      0.0
3                   3509         PayPal            37      0.0
4                   3452    Credit Card            49      0.0

         Customer Name  Age  Gender  Churn
0  Christine Hernandez   37    Male      0
1  Christine Hernandez   37    Male      0
2  Christine Hernandez   37    Male      0
```

```
3  Christine Hernandez   37    Male      0
4          James Grant   49  Female      1
```

```
print(df1.columns)
print(df2.columns)
```

```
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
       'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',
       'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
      dtype='object')
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
       'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',
       'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
      dtype='object')
```

```
import pandas as pd

df1 = pd.read_csv('/content/ecommerce_customer_data_large.csv')

df2 = pd.read_csv('/content/ecommerce_customer_data_custom_ratios.csv')

df = pd.concat([df1, df2], ignore_index=True)

print(df.shape)
```

```
(500000, 13)
```

```
df.head()
```

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Custom A
0	44605	2023-05-03 21:30:02	Home	177	1	2427	PayPal	
1	44605	2021-05-16 13:57:44	Electronics	174	3	2448	PayPal	
2	44605	2020-07-13 06:16:57	Books	413	1	2345	Credit Card	
3	44605	2023-01-17	Electronics	396	3	937	Cash	

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500000 entries, 0 to 499999
Data columns (total 13 columns):
 #   Column                 Non-Null Count   Dtype
```

```
---  ------                 --------------   -----
 0   Customer ID            500000 non-null  int64
 1   Purchase Date          500000 non-null  object
 2   Product Category       500000 non-null  object
 3   Product Price          500000 non-null  int64
 4   Quantity               500000 non-null  int64
 5   Total Purchase Amount  500000 non-null  int64
 6   Payment Method         500000 non-null  object
 7   Customer Age           500000 non-null  int64
 8   Returns                405022 non-null  float64
 9   Customer Name          500000 non-null  object
 10  Age                    500000 non-null  int64
 11  Gender                 500000 non-null  object
 12  Churn                  500000 non-null  int64
dtypes: float64(1), int64(7), object(5)
memory usage: 49.6+ MB
```

🧹 Data Cleaning

```
df.isnull().sum()
```

	0
Customer ID	0
Purchase Date	0
Product Category	0
Product Price	0
Quantity	0
Total Purchase Amount	0
Payment Method	0
Customer Age	0
Returns	94978
Customer Name	0
Age	0
Gender	0
Churn	0

dtype: int64

```
df.drop_duplicates(inplace=True)

df.dropna(inplace=True)

df.columns = df.columns.str.strip()
```

```
df['Total Purchase Amount']
```

	Total Purchase Amount
0	2427
1	2448
2	2345
3	937
4	2598
...	...
499995	2187
499996	3615
499997	2466
499998	3668
499999	2370

405022 rows × 1 columns

dtype: int64

```
df['Purchase Date'] = pd.to_datetime(df['Purchase Date'])
```

```
df['Month'] = df['Purchase Date'].dt.month
```

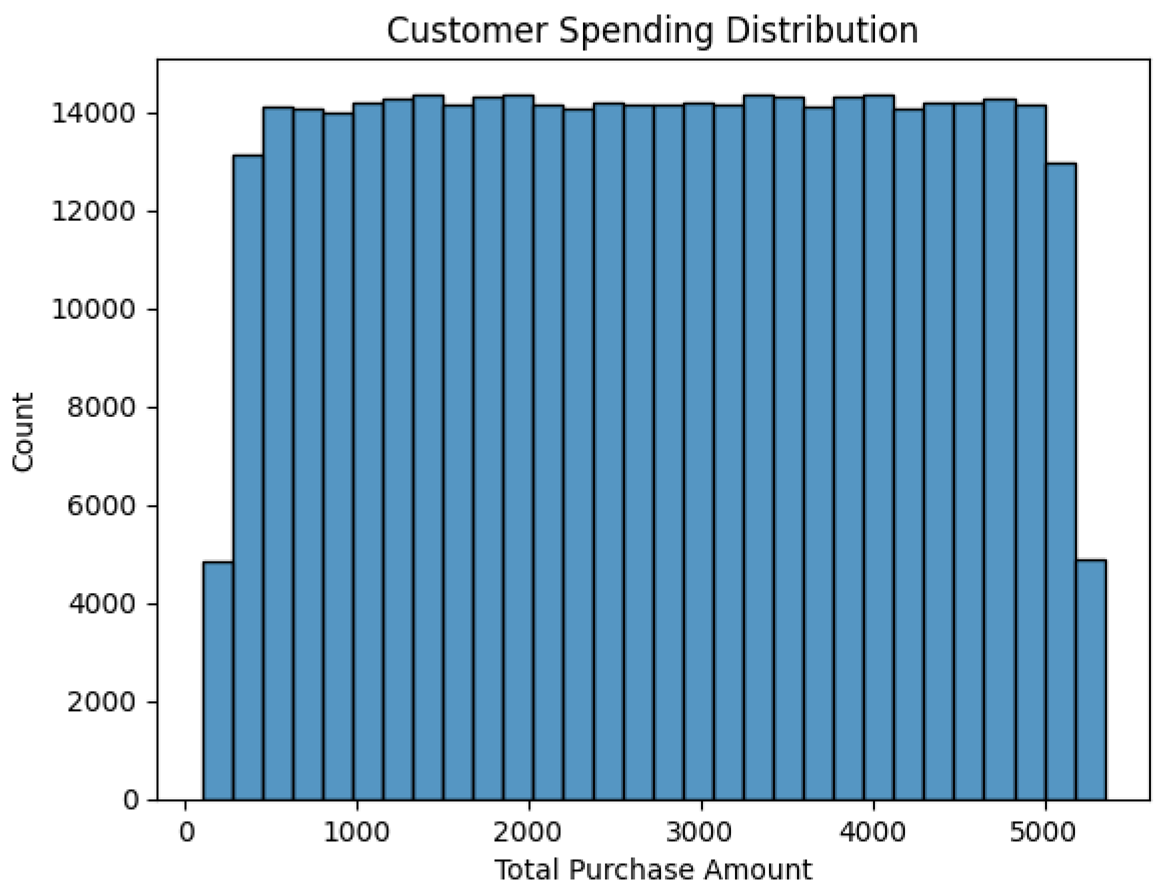
📊 Exploratory Data Analysis

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.histplot(df['Total Purchase Amount'], bins=30)

plt.title('Customer Spending Distribution')

plt.show()
```



Spending Distribution Graph Insights

Insights

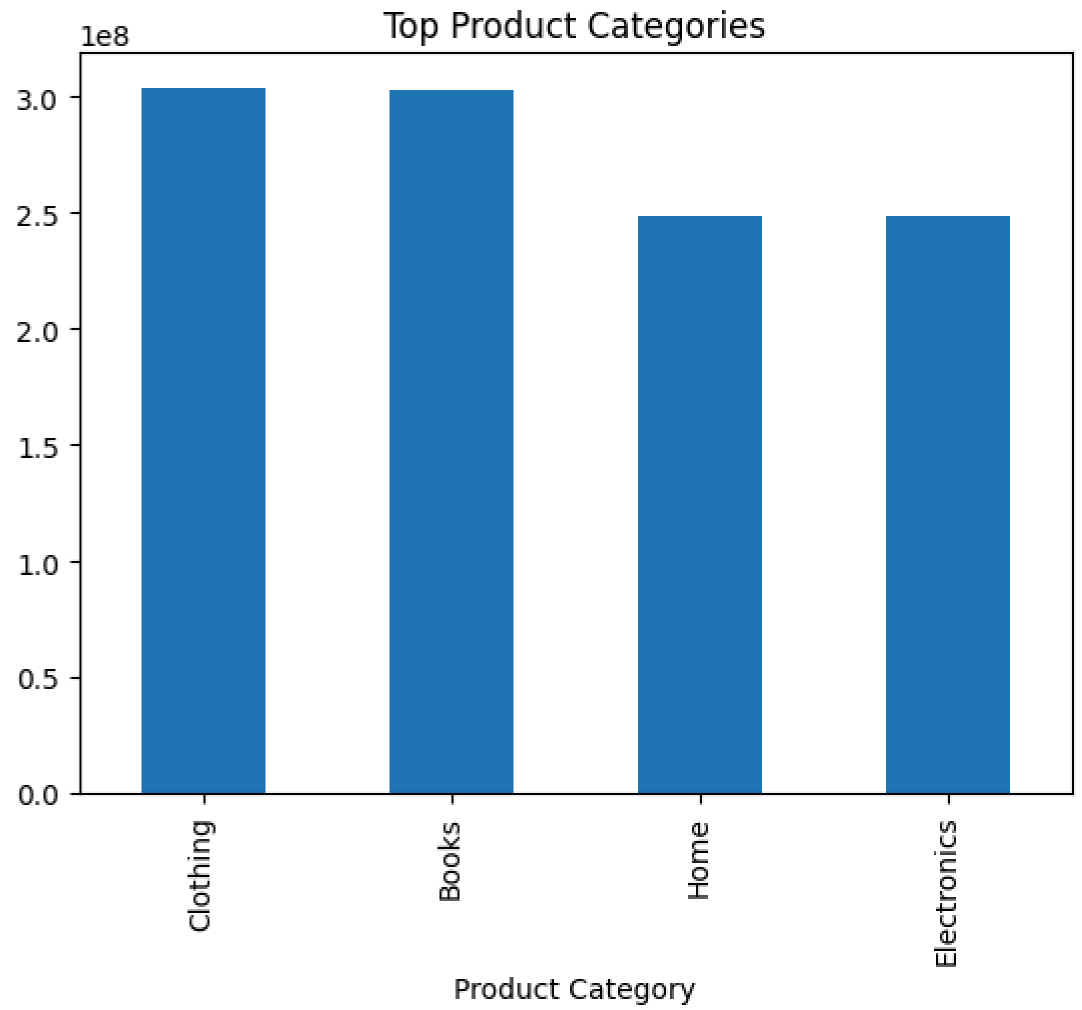
- Most customers spend moderate amounts on purchases.
- A smaller group of customers contributes significantly higher revenue.
- The spending distribution is positively skewed, indicating the presence of high-value customers.
- These high-spending customers can be targeted with premium offers and loyalty programs.

```
top_products = df.groupby('Product Category')['Total Purchase Amount'].sum()

top_products.plot(kind='bar')

plt.title('Top Product Categories')

plt.show()
```



📈🛍️ Top Product Categories Graph Insights

Insights

- Certain product categories generate significantly higher revenue than others.
- High-performing categories should receive more marketing focus and inventory support.
- Low-performing categories may require promotional campaigns or pricing adjustments.
- Understanding product demand helps improve business strategy and sales optimization.

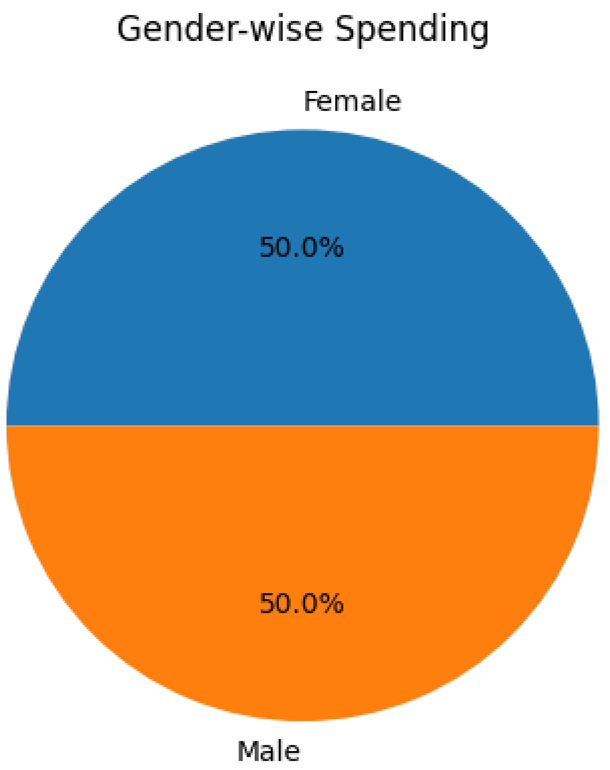
```
gender_sales = df.groupby('Gender')['Total Purchase Amount'].sum()

gender_sales.plot(kind='pie', autopct='%1.1f%%')

plt.ylabel('')

plt.title('Gender-wise Spending')
```

```
plt.show()
```



⚖️ 📊 ♀️ ♂️ Gender-wise Spending Insights

Insights

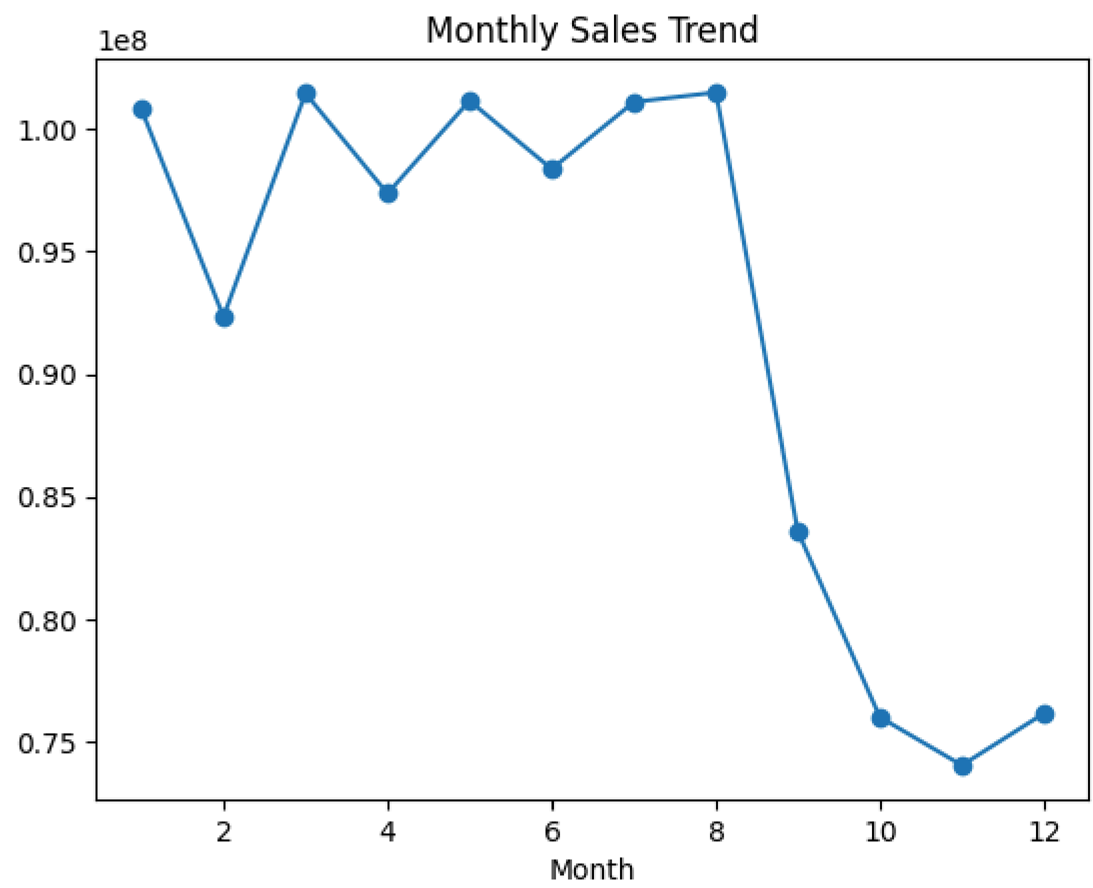
- Spending behavior varies across gender groups.
- One gender group contributes a larger share of total revenue.
- Personalized marketing campaigns can improve customer engagement.
- Gender-based customer preferences can help optimize product recommendations.

```
monthly_sales = df.groupby('Month')['Total Purchase Amount'].sum()

monthly_sales.plot(marker='o')

plt.title('Monthly Sales Trend')

plt.show()
```



** 📈 📉 📊 Monthly Sales Trend Insights**

Insights

- Sales fluctuate across different months, indicating seasonal purchasing behavior.
- Certain months show peak customer activity and higher revenue generation.
- Businesses can increase marketing efforts during high-sales periods.
- Lower-sales months may require promotional campaigns to improve engagement.

```
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

```
features = df[['Total Purchase Amount', 'Quantity']]
```

```
scaler = StandardScaler()

scaled_features = scaler.fit_transform(features)
```

```
kmeans = KMeans(n_clusters=3, random_state=42)
```

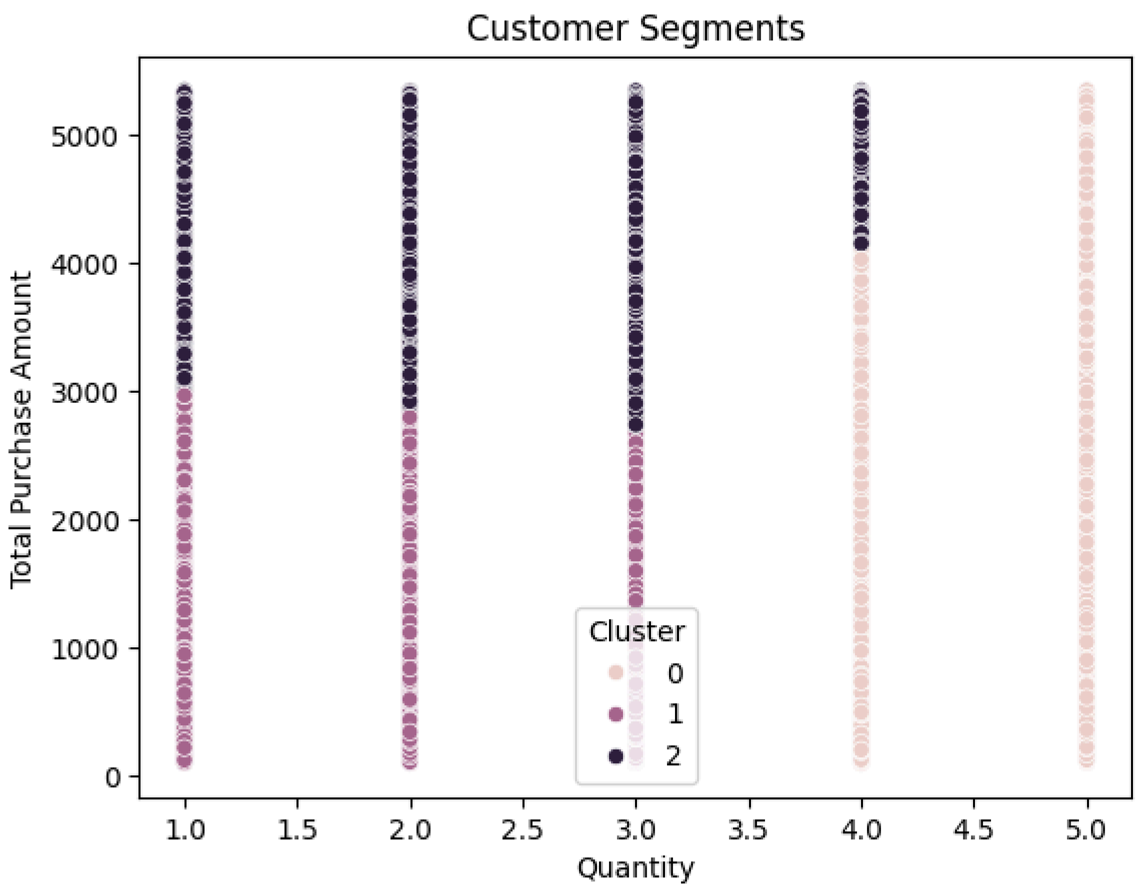
```
df['Cluster'] = kmeans.fit_predict(scaled_features)
```

```
sns.scatterplot(
    x=df['Quantity'],
    y=df['Total Purchase Amount'],
    hue=df['Cluster']
)

plt.title('Customer Segments')

plt.show()
```

```
/usr/local/lib/python3.12/dist-packages/IPython/core/pylabtools.py:151: UserW
  fig.canvas.print_figure(bytes_io, **kw)
```



📊🔍 Customer Segmentation Insights

Insights

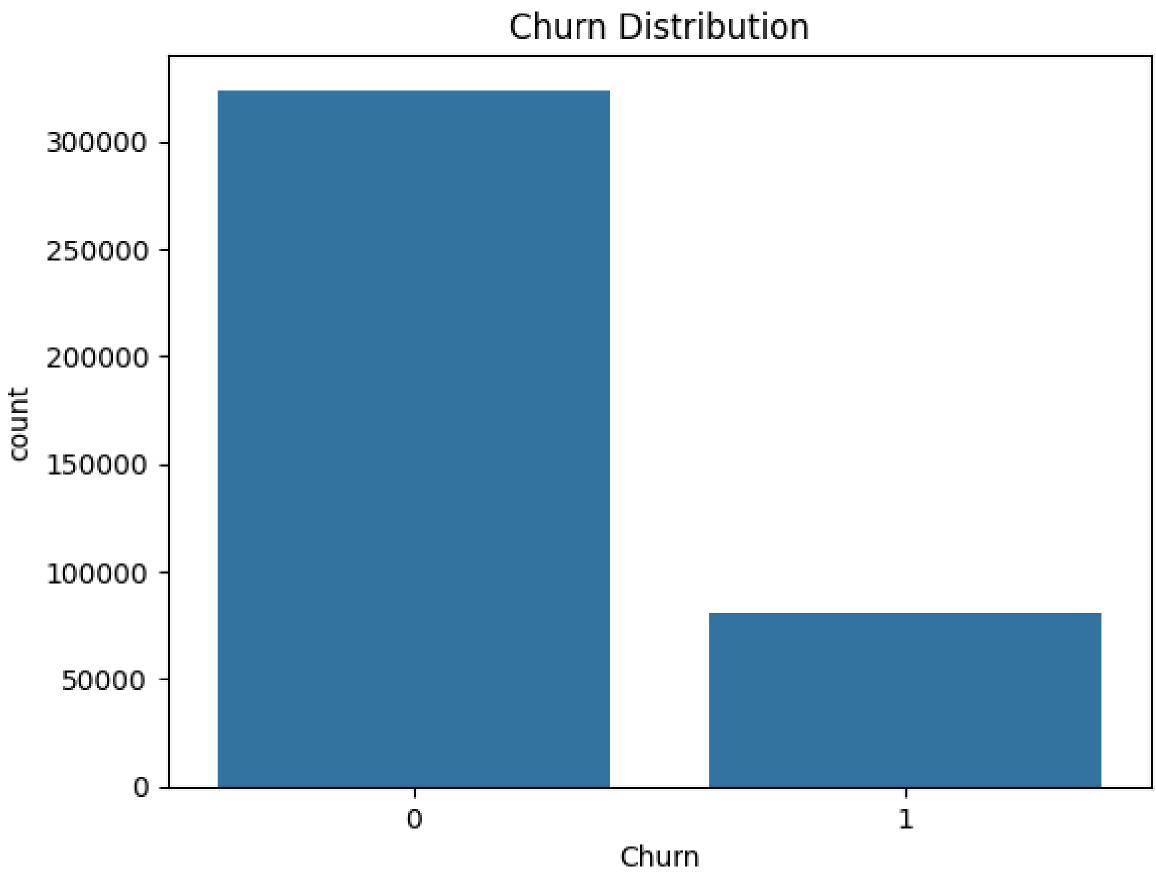
- Customers were successfully grouped into different spending segments.
- One cluster represents high-value customers with larger purchase amounts.
- Another cluster contains low-spending customers with lower engagement.
- Segment-specific marketing strategies can improve customer retention and revenue.

- High-value customers should be prioritized through loyalty rewards and personalized services.

```
sns.countplot(x='Churn', data=df)

plt.title('Churn Distribution')

plt.show()
```



** 📊 📉 Churn Distribution Insights**

Insights

- A noticeable percentage of customers are marked as churned.
- Churned customers indicate potential customer retention issues.
- Businesses should focus on re-engagement strategies to reduce churn.
- Personalized discounts and loyalty programs may help retain at-risk customers.
- Reducing churn can significantly improve long-term profitability.

```
churn_percent = df['Churn'].value_counts(normalize=True) * 100

print(churn_percent)
```

```
Churn
0    79.974421
1    20.025579
Name: proportion, dtype: float64
```

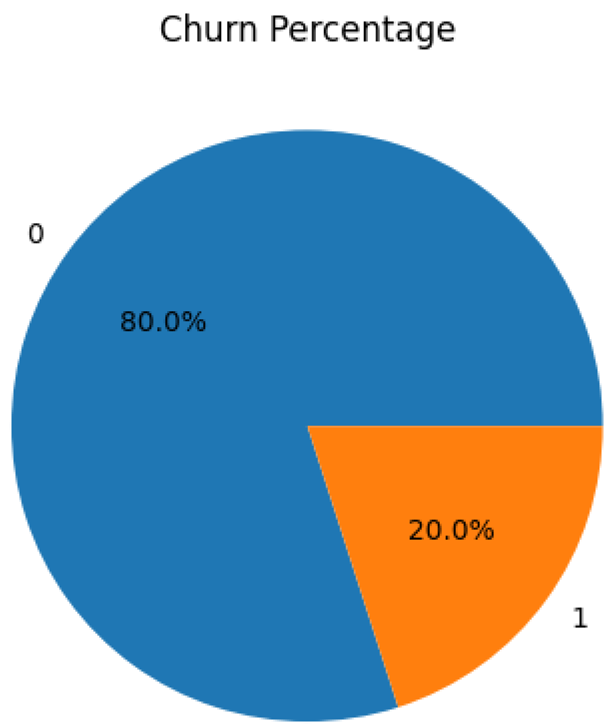
```
import matplotlib.pyplot as plt

churn_percent.plot(kind='pie', autopct='%1.1f%%')

plt.ylabel('')

plt.title('Churn Percentage')

plt.show()
```



🔍 📊 📉 Churn Analysis Insights

A significant percentage of customers are marked as churned. This indicates potential customer retention issues.

Customers with churn risk should be targeted with:

personalized offers loyalty rewards re-engagement campaigns Reducing churn can improve long-term revenue and customer satisfaction.

Key Findings

- High-value customer groups were identified using clustering techniques.
- Customer purchasing behavior varies across different segments.
- Seasonal trends influence overall sales performance.
- Churn analysis revealed customers at risk of disengagement.
- Product category analysis identified high-performing products.
- Data-driven marketing strategies can improve customer retention and revenue growth.

Recommendations

1. Introduce loyalty programs for repeat and high-value customers.
2. Provide personalized product recommendations based on purchase history.
3. Offer discounts and retention campaigns for churn-risk customers.
4. Increase marketing efforts during peak sales periods.
5. Focus inventory and promotions on high-performing product categories.

Conclusion

This project successfully analyzed customer purchasing behavior using data analytics and visualization techniques.

The analysis identified important purchasing trends, customer segments, high-value customers, and churn risks.